

		the 20 °C
14	B	For an ideal gas, $pV = nRT$ thus temperature is directly proportional to the product of pressure and volume. From L to M, pressure stays constant, thus temperature will increase linearly with volume. Thus, the only option is option B. For M to N, $p_M V_M = 2.0 \times 10^6 \times 0.003 = 6000 \text{ J}$ $p_L V_L = 0.8 \times 10^6 \times 0.005 = 4000 \text{ J}$ Since the product of pressure and volume at state M is larger than at state L, the temperature is higher at M than at L.
15	B	ΔU increase in internal energy of system